

PRACTICAL COURSE WORKBOOK

SQL for Financial Analysis

Model, query and explain dependable data

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	SQL, Finance, Analytics, Window functions, Reporting, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use SQL appropriately in a realistic, bounded task.
- Use Finance appropriately in a realistic, bounded task.
- Use Analytics appropriately in a realistic, bounded task.
- Use Window functions appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

SQL for Financial Analysis is a structured, practical course covering SQL, Finance, Analytics, Window functions, Reporting. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 SQL: Data modelling	1. Entities with SQL 2. Relationships with Finance 3. Constraints with Analytics
02 Finance: Querying	1. Selection with Window functions 2. Joins with Reporting 3. Aggregation with SQL
03 Analytics: Reliability	1. Transactions with Finance 2. Indexes with Analytics 3. Backups with Window functions
04 Window functions: Production design	1. Security with Reporting 2. Performance with SQL 3. Operations with Finance

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

SQL: Data modelling

Apply SQL through entities, relationships, constraints.

LESSON 01 / 18 MIN

Entities with SQL

Learning objectives

- Explain entities in the context of SQL for Financial Analysis.
- Apply SQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

entities, SQL, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a entities result produced with SQL?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Relationships with Finance

Learning objectives

- Explain relationships in the context of SQL for Financial Analysis.
- Apply Finance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

relationships, Finance, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a relationships result produced with Finance?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Constraints with Analytics

Learning objectives

- Explain constraints in the context of SQL for Financial Analysis.
- Apply Analytics to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

constraints, Analytics, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a constraints result produced with Analytics?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Finance: Querying

Apply Finance through selection, joins, aggregation.

LESSON 04 / 30 MIN

Selection with Window functions

Learning objectives

- Explain selection in the context of SQL for Financial Analysis.
- Apply Window functions to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

selection, Window functions, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a selection result produced with Window functions?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Joins with Reporting

Learning objectives

- Explain joins in the context of SQL for Financial Analysis.
- Apply Reporting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

joins, Reporting, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a joins result produced with Reporting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Aggregation with SQL

Learning objectives

- Explain aggregation in the context of SQL for Financial Analysis.
- Apply SQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

aggregation, SQL, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a aggregation result produced with SQL?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Analytics: Reliability

Apply Analytics through transactions, indexes, backups.

LESSON 07 / 26 MIN

Transactions with Finance

Learning objectives

- Explain transactions in the context of SQL for Financial Analysis.
- Apply Finance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transactions, Finance, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a transactions result produced with Finance?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Indexes with Analytics

Learning objectives

- Explain indexes in the context of SQL for Financial Analysis.
- Apply Analytics to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

indexes, Analytics, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a indexes result produced with Analytics?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Backups with Window functions

Learning objectives

- Explain backups in the context of SQL for Financial Analysis.
- Apply Window functions to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

backups, Window functions, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a backups result produced with Window functions?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Window functions: Production design

Apply Window functions through security, performance, operations.

LESSON 10 / 22 MIN

Security with Reporting

Learning objectives

- Explain security in the context of SQL for Financial Analysis.
- Apply Reporting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, Reporting, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a security result produced with Reporting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Performance with SQL

Learning objectives

- Explain performance in the context of SQL for Financial Analysis.
- Apply SQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

performance, SQL, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a performance result produced with SQL?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Operations with Finance

Learning objectives

- Explain operations in the context of SQL for Financial Analysis.
- Apply Finance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

operations, Finance, databases

Retrieval prompt

In SQL for Financial Analysis, which evidence best supports a operations result produced with Finance?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable SQL for Financial Analysis project using SQL, Finance, Analytics.

DELIVERABLE	A working SQL for Financial Analysis artefact plus an evidence-based self-review.
TOOLS	A suitable SQL environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using SQL.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21

<https://www.postgresql.org/docs/current/tutorial.html>

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21

<https://www.postgresql.org/docs/current/sql.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the SQL scope.